

MERN + JavaScript Interview Questions with Answers

1. What is JavaScript?

Answer: JavaScript is a programming language used to create dynamic and interactive web pages. It runs in browsers and also on the server using Node.js.

2. Difference between var, let, const

Answer: var is function scoped, let and const are block scoped. const cannot be reassigned.

3. Difference between == and ===

Answer: == compares value with type conversion while === compares value and type.

4. What is closure?

Answer: A closure is when a function remembers variables from its outer scope even after the outer function has finished executing.

5. What is Promise?

Answer: A Promise represents the result of an asynchronous operation and has states: pending, fulfilled, rejected.

6. What is async/await?

Answer: A modern way to handle asynchronous code built on top of Promises.

7. What is the event loop?

Answer: The event loop allows JavaScript to perform non-blocking operations by handling asynchronous callbacks.

8. What is hoisting?

Answer: JavaScript moves variable and function declarations to the top of their scope during execution.

9. What are arrow functions?

Answer: Short syntax for writing functions using =>.

10. Difference between synchronous and asynchronous JS

Answer: Synchronous runs line by line while asynchronous allows tasks to run without blocking execution.

11. What is React?

Answer: React is a JavaScript library for building user interfaces using reusable components.

12. What is JSX?

Answer: JSX allows writing HTML-like syntax inside JavaScript.

13. What is Virtual DOM?

Answer: A lightweight copy of the real DOM used by React to update only changed elements.

14. Difference between state and props

Answer: State is local and mutable while props are passed from parent and read-only.

15. What are React Hooks?

Answer: Functions that allow functional components to use state and lifecycle features.

16. What is useState?

Answer: A hook used to manage state in functional components.

17. What is useEffect?

Answer: A hook used for side effects like API calls or timers.

18. What is React Router?

Answer: A library used to handle navigation between pages in React apps.

19. What is lifting state up?

Answer: Moving state to a common parent component so multiple components can access it.

20. Controlled vs uncontrolled components

Answer: Controlled components manage form data using React state, uncontrolled use the DOM.

21. What is Node.js?

Answer: Node.js is a runtime environment that allows JavaScript to run on the server.

22. Event-driven architecture in Node.js

Answer: Node handles operations through events and callbacks.

23. What is npm?

Answer: Node Package Manager used to install and manage packages.

24. What is package.json?

Answer: File containing project metadata, dependencies, and scripts.

25. What are environment variables?

Answer: Variables used to store sensitive configuration like API keys or DB URLs.

26. What is middleware?

Answer: Functions executed between request and response in Express.

27. What is Express.js?

Answer: A minimal Node.js web framework used to build APIs and servers.

28. What is routing?

Answer: Routing determines how an application responds to client requests.

29. HTTP methods

Answer: GET retrieve data, POST create, PUT update, DELETE remove.

30. HTTP status codes

Answer: Codes such as 200 success, 404 not found, 500 server error.

31. What is CORS?

Answer: Cross-Origin Resource Sharing allows requests from different domains.

32. Handling errors in Express

Answer: Using error-handling middleware.

33. What is MongoDB?

Answer: A NoSQL database that stores data in JSON-like documents.

34. What is a document?

Answer: A single record stored in BSON format.

35. What is a collection?

Answer: A group of MongoDB documents similar to a SQL table.

36. SQL vs MongoDB

Answer: SQL uses tables and fixed schema while MongoDB uses collections and flexible schema.

37. What is indexing?

Answer: Indexes improve database query performance.

38. What is aggregation?

Answer: Used to process and analyze data using pipelines.

39. What is Mongoose?

Answer: An ODM library used with Node.js to interact with MongoDB.

40. What does MERN stand for?

Answer: MongoDB, Express.js, React, Node.js.

41. How MERN works

Answer: React frontend communicates with Node/Express backend which interacts with MongoDB.

42. What is REST API?

Answer: A way for systems to communicate using HTTP endpoints.

43. Connecting React with Node backend

Answer: Using API calls via fetch or axios.

44. What is JWT?

Answer: JSON Web Token used for authentication.

45. What is bcrypt?

Answer: Library used to hash passwords securely.

46. What is MVC architecture?

Answer: Model handles data, View UI, Controller logic.

47. What is pagination?

Answer: Dividing large datasets into smaller pages.

48. What is debouncing?

Answer: Limiting how often a function executes.

49. Deploy MERN app

Answer: Frontend on Vercel/Netlify, backend on Render/AWS, database on MongoDB Atlas.

50. Explain a MERN project

Answer: Candidate describes project features, tech stack, and challenges.